

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Laws of Logarithms

Logarithm questions reward one clean compression step before solving.

LAWS OF LOGARITHMS · P2 · 09

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DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 LAWS: Compress Logs

TRIGGER *sum or difference of logs*

02 SOLVE: Solve Log Equation

TRIGGER *log equation*

03 BASE: Change Base

TRIGGER *different bases*

04 EXP: Log To Exponential

TRIGGER *definition of log*

BANK EVIDENCE Local pages: 17 problem, 40 answer, 17 notes. Website primary entries: 20.

01 LAWS: Compress Logs

TRIGGER *sum or difference of logs*

BECOMES Use log laws to combine into one log.

FIRST LINE TO WRITE

$$\log a + \log b = \log(ab)$$

SIMPLEST STRATEGY

- 1 Look for plus, minus, or multiplier.
- 2 Apply product, quotient, or power law.
- 3 Write one log.
- 4 Solve or simplify.

WORKED MODEL

$$\log x + \log(x - 3) = \log(x(x - 3)).$$

HARD VARIANTS

- 1 All logs in a law must have the same base.
- 2 Check log arguments are positive after solving.
- 3 Move coefficients into powers before combining.

— BOTTOM LINE

One log is easier to solve than many.

02 SOLVE: Solve Log Equation

TRIGGER *log equation*

BECOMES Compress, then remove the log.

FIRST LINE TO WRITE

$$\log_a F(x) = \log_a G(x) \Rightarrow F(x) = G(x)$$

SIMPLEST STRATEGY

- 1 Same base confirmed.
- 2 One log each side.
- 3 Lift the log.
- 4 Verify domain.
- 5 End with valid roots.

WORKED MODEL

$$\log x + \log(x - 3) = 1 \Rightarrow x(x - 3) = 10.$$

HARD VARIANTS

- 1 If one side is a number, rewrite it as a log of a power.
- 2 Quadratics from logs can create invalid negative arguments.
- 3 Never take log of zero or negative values.

— BOTTOM LINE

Log equations always finish with a domain check.

03 BASE: Change Base

TRIGGER *different bases*

BECOMES Rewrite using a common base or change-base formula.

FIRST LINE TO WRITE

$$\log_a b = \frac{\log b}{\log a}$$

SIMPLEST STRATEGY

- 1 Bring bases into view.
- 2 Align with powers if possible.
- 3 Switch base if needed.
- 4 Evaluate or solve.

WORKED MODEL

$$\log_4 x = \frac{1}{2} \log_2 x.$$

HARD VARIANTS

- 1 If bases are powers of the same number, rewrite directly.
- 2 In equations, convert every log to the same base.
- 3 Keep exact logarithmic form unless decimals are requested.

— BOTTOM LINE

Changing base turns unfamiliar logs into familiar ones.

04 EXP: Log To Exponential

TRIGGER *definition of log*

BECOMES Translate between log and power form.

FIRST LINE TO WRITE

$$\log_a x = b \Rightarrow x = a^b$$

SIMPLEST STRATEGY

- 1 Extract base.
- 2 Xchange log form for exponent form.
- 3 Place the argument as the result.

WORKED MODEL

$$\log_3 x = 4 \Rightarrow x = 81.$$

HARD VARIANTS

- 1 Use exponential form after compressing logs.
- 2 Check the base is positive and not 1.
- 3 Check the argument is positive in the original equation.

— BOTTOM LINE

A logarithm is an exponent.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
sum or difference of logs	$\log a + \log b = \log(ab)$
log equation	$\log_a F(x) = \log_a G(x) \Rightarrow F(x) = G(x)$
different bases	$\log_a b = \frac{\log b}{\log a}$
definition of log	$\log_a x = b \Rightarrow x = a^b$